

5 Haemophilia is a group of rare conditions that affects the ability of the blood to clot.

These conditions are caused by mutations in the genes coding for factors involved in the blood clotting process.

- (a) Haemophilia is inherited in a similar way to red-green colour blindness and most people who have it are male.

Explain why most people who have haemophilia are male.

(4)

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- (b) Two types of haemophilia are haemophilia A and haemophilia B.

In 2021, there were 7.87×10^9 people in the world.

There were 185 218 people with haemophilia A and 37 998 people with haemophilia B.

- (i) Calculate the ratio of people with haemophilia A to those with haemophilia B.
Give your answer to **two** significant figures.

(2)

Ratio : 1



(ii) Which is the percentage of the world population who had haemophilia A, in 2021?

(1)

- ☐ **A** 2.35×10^5
- ☐ **B** 2.35×10^3
- ☐ **C** 2.35×10^{-3}
- ☐ **D** 2.35×10^{-5}

(iii) The number of people with haemophilia is only an estimate.

Give **one** reason why this is only an estimate.

(1)

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(c) In haemophilia, activation of the blood clotting pathway results in less thrombin being produced than in healthy individuals.

Explain how reduced thrombin will affect blood clotting.

(3)

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(Total for Question 5 = 11 marks)